

generations overlap at any given time. A demographic shock changes the composition of a given cohort relative to other cohorts through such events as war (like World Wars I and II), a baby boom (like the generation born in the two decades following World War II), or infectious disease (Spanish Flu in 1918).

Several OLG models predict that demographic composition affects expected returns. Theory suggests two main avenues for this to occur. First, the life-cycle smoothing in the OLG framework requires that when the middle-aged to young population is small, there is excess demand for consumption by a relatively large cohort of retirees. Retirees do not want to hold financial assets: in fact, they are selling them to fund their consumption. For markets to clear, asset prices have to fall.¹⁶ Abel (2001) uses this intuition to predict that as baby boomers retire, stock prices will decline. The predictions are not, however, clear cut: Brooks (2002), for example, argues that the baby boom effect on asset prices is weak. The second mechanism where demography can predict stock returns is that, since different cohorts have different risk characteristics, asset prices change as the aggregate risk characteristics of the economy change. In an influential study, Bakshi and Chen (1994) show that risk aversion increases as people age and, as the average age rises in the population, the equity premium should increase.

In testing a link between demographic risk and asset returns, it is important to use international data; using only one country's demographic experience is highly suspect because demographic changes are so gradual. The literature employing cross-country analysis includes Erb, Harvey, and Viskanta (1997), Ang and Maddaloni (2005), and Arnott and Chaves (2011). There is compelling international empirical evidence that demography does affect risk premiums.

Political Risk

The last macro risk that an asset owner should consider is *political* or *sovereign risk*. Political risk has been always important in emerging markets: the greater the political risk, the higher the risk premiums required to compensate investors for bearing it. Political risk was thought to be of concern only in emerging markets.¹⁷ The financial crisis changed this, and going forward political risk will also be important in developed countries.¹⁸

3. Dynamic Factors

The CAPM factor is the market portfolio, and, with low-cost index funds, exchange-traded funds, and stock futures, the market factor is tradeable. Other

¹⁶This is the main economic mechanism in, for example, Geanakoplos, Magill, and Quinzii (2004).

¹⁷Harvey (2004) finds little evidence, for example, that political risk is reflected in developed countries.

¹⁸For a recent paper showing how political risk affects equity risk premiums, see Pástor and Veronesi (2012).